Md. Jubayer Hossain

Bioinformatics & Computational Biology

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EDUCATION

Master of Science in Microbiology, Jagannath University, Dhaka, Bangladesh 2019 – 2020

Bachelor of Science in Microbiology, Jagannath University, Dhaka, Bangladesh 2016 – 2019

RESEARCH EXPERIENCE

Founder & Executive Director, CHIRAL Bangladesh (Institute for Volunteering Research), Dhaka, Bangladesh Jun 2020 – Present

Research Collaborator, Yeon Ju Kim, Ph.D., Kyung Hee University, Seoul, South Korea Sep 2026 – Present

Single-Cell Transcriptomic Identification of Alzheimer's Disease Associated Molecular Targets and CNS–Peripheral Immune Signatures for Experimental Validation and Therapeutic Intervention

Single-Cell Multi-Organ Profiling to In Vivo Validation: Identification of Hyperglycemia-Associated Gene Signatures and Their Modulation by GV1 in ObT2D Mice

Research Collaborator, Iqbal Mahmud, Ph.D., Lamar University, Beaumont, TX, USA 2026 – Present

EWSAtlas: A Harmonized Single-Cell Transcriptomic Atlas of Ewing Sarcoma

Unified Single-Cell Atlas of Diffuse Midline Glioma (DMG)

Research Collaborator, Md. Salequl Islam, Ph.D., One Health Lab, Jahangirnagar University, Dhaka, Bangladesh 2024 – Present

Genomic Profiling of Extended-Spectrum Antimicrobial Resistance Genes (ARGs) and Mobilizable Resistance Determinants in Industrial Effluents

One Health Assessment of Antimicrobial Usage, Microbial Dynamics, and Resistance Transmission Pathways Across Poultry Farm Environments and Farm Workers in Bangladesh

PROFESSIONAL EXPERIENCE

Founder & CEO, DeepBio Limited Jan 2025 – Present

Faculty (Bioinformatics), cBLAST, University of Dhaka Aug 2023 – Present

Program Lead, GSA Bioinformatics Internship Jul 2025 – Present

Curriculum Developer and Instructor, School of Bioinformatics Jan 2024 – Present

Teaching Assistant, Department of Microbiology, Jagannath University Oct 2022 – Dec 2022

SELECTED PUBLICATIONS

Underline indicates mentee; * corresponding

1. Islam, M. W., Rahman, M. M., Naznin, H., Hossain, M. S., Akter, T., Shatabde, Z. A., & ***Hossain, M. J.** (2026) "Integrative Bioinformatics Analysis Reveals COL13A1 and COL23A1 as Potential Diagnostic and Prognostic Biomarkers in Thyroid Cancer" *Health Science Reports*. DOI: 10.1002/hsr2.72617
2. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & ***Hossain, M. J.** (2025) "Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer" *Scientific Reports*. DOI: 10.1038/s41598-025-13315-6
3. Ahmed, M. Z., Billah, M. M., Ferdous, J., & ***Hossain, M. J.** (2025) "Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors" *Scientific Reports*. DOI: 10.1038/s41598-025-88339-z
4. Bari, S. M., Fuad, M., **Hossain, M. J.**, Ahammad, I., Begum, M., & Helal, M. M. U. (2025) "A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout" *BMC Genomics*. DOI: 10.1186/s12864-025-12127-2
5. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & ***Hossain, M. J.** (2026) "Computational identification of Taxus baccata-derived phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma" *Scientific Reports* (Under Review). DOI: 10.21203/rs.3.rs-8494315/v1
6. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & ***Hossain, M. J.** (2025) "Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer" *Health Science Reports* (Under Review).

7. Billah, M. M., Mabsurah, K., Ahammad, K., Yeana, I. J., Sumaiya, M., Islam, T., Bhattacharjee, A., Ferdous, J., & *__Hossain, M. J.__ (2025) "Multi-omics pan-cancer analysis reveals an immunological role and prognostic potential of WDR76" *BMC Genomics* (Under Review).
8. **Hossain, M. J.** (2026) "metaXpress: An End-to-End Bioconductor Framework for Quality Control, Harmonization, and Multi-Cohort Bulk RNA-seq Meta-Analysis" *bioRxiv*. GitHub
9. **Hossain, M. J.** (2026) "Graph Neural Network for predicting immunosuppressive spatial niches in glioblastoma from 10x Visium spatial transcriptomics" *bioRxiv*. GitHub
10. **Hossain, M. J.** (2026) "Translating Perturb-seq Screens Across Cell Lines" Preprint. GitHub
11. **Hossain, M. J.**, & Dev, P.C. (2026) "Evaluating the false-positive rate of single-cell copy-number and malignant-cell classifiers" *bioRxiv*. GitHub
12. **Hossain, M. J.**, Dev, P. C., Nawar, T., Arony, T. A., Hoque, M., Chowdhury, D., & Rahman, M. A. (2026) "A donor-level map of immune sex differences" *bioRxiv*. GitHub

SELECTED PRESENTATIONS

1. Rahman, M. A., Shahjahan, M. and *__Hossain, M. J.__ EBV-antigen-specific T cell clonotype fate in nasopharyngeal carcinoma: A single-cell TCR analysis. Dhaka University Research Society (DURS), December, 2026.
2. Chowdhury, D., Shahjahan, M. and *__Hossain, M. J.__ Reassessing the chemotherapy–TIGIT association in pancreatic cancer through cross-cohort validation. Dhaka University Research Society (DURS), December, 2026.
3. Hoque, M., Shahjahan, M. and *__Hossain, M. J.__ Chemotherapy-induced tumor cell state dynamics in osteosarcoma: NMF-based decomposition and independent bulk validation. Dhaka University Research Society (DURS), December, 2026.
4. Arony, T. A., Shahjahan, M. and *__Hossain, M. J.__ Spatial mapping of diffuse midline glioma cell states reveals conserved territorial architecture and a mesenchymal-like interface for tumor–microenvironment crosstalk. Dhaka University Research Society (DURS), December, 2026.
5. Nawar, T., Shahjahan, M., *__Hossain, M. J.__ Cross-Cohort Meta-Analysis of Acral Melanoma Immune Cell Quantity vs. Function With Three-Way Decomposition. Dhaka University Research Society (DURS), December, 2026.
6. Islam Abhi, S. N., Meme, M. A., Shahariar, M., Barsha, L., *__Hossain, M. J.__ Single-cell transcriptomic and network pharmacology insights into the therapeutic potential of Withania somnifera-derived phytochemicals against cisplatin-resistant ovarian cancer. BEICTCON2026, IEEE, August, 2026.
7. Islam, M. M., *__Hossain, M. J.__ Fezf2-Mediated Cortical Development: Multi-Omics Single-Cell Analysis. Network of Young Biotechnologists Bangladesh, April, 2026.
8. Ahmad, F., Fuad, M., Shahjahan, M. and *__Hossain, M. J.__ Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis. Bangladesh Society of Microbiologists, December, 2025.

AWARDS & CERTIFICATIONS

Open Bioinformatics Foundation Event Fellowship, *Open Bioinformatics Foundation* 2026
Travel Grant for scverse x cellpainting Hackathon, *scverse* 2026
Scholarship for AI in Public Health Workshop, *Bill and Melinda Gates Foundation* 2023
Dutch-Bangla Bank (DBBL) Merit Scholarship, *Dutch-Bangla Bank* 2013
scverse × Cell Painting Hackathon, scverse Consortium 2026
Interdisciplinary Computational Biology, BRAC University 2026

TECHNICAL SKILLS

Bioinformatics: Bulk RNA-seq, single-cell RNA-seq, spatial transcriptomics, multi-omics integration, WGS, metagenomics
Programming & ML: Python, R, Julia, SQL, Bash, JavaScript; scikit-learn, PyTorch, RDKit, DeepChem
Workflows, HPC & Cloud: Nextflow (DSL2), nf-core, Docker, Singularity, Git; SLURM; AWS S3, Google Cloud Storage